

# ELE8101: Estimation Theory

## Coursework 2 — Vehicle State Estimation on a Racing Track

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### Abstract

This report addresses the estimation of the position and velocity of a racing vehicle on an asymmetric teardrop track from noisy range-only beacon measurements. A bound-respecting nonlinear vehicle motion model with smooth  $C^\infty$  saturations is used as the truth simulator; an optimised five-beacon GDOP-minimising layout attenuates ranging noise across the entire track. An Extended Kalman Filter is implemented with prediction-step Jacobians obtained from CasADi automatic differentiation at every step, ensuring the predictor and the simulator share identical dynamics. Across 100 Monte-Carlo trials the filter achieves a mean arc-length error of 0.226 m, well below the  $\sigma_y \overline{\text{GDOP}} = 1.379$  m single-measurement GDOP scale, and passes a Normalized Innovation Squared consistency check (empirical mean 4.994, 95% of samples in the 95% confidence band). A Moving Horizon Estimator with the same nonlinear constraint exceeds the 10 ms sampling budget by a factor of 128, supporting the EKF as the preferred estimator for this implementation. A 9-state augmented EKF recovers injected sensor biases up to 0.8 m to within 0.04 m. Source code and `results_summary.json` are available at [github.com/cuichenhui9-eng/ELE8101-CW2](https://github.com/cuichenhui9-eng/ELE8101-CW2).

## 1 System Modelling

### 1.1 Track Geometry

The track has an asymmetric teardrop shape comprising two circular arcs connected by two inclined common external tangent straights. Arc  $A$  (large) has centre  $A$  and centreline radius  $R_A = (R + r)/2 = 48$  m; arc  $B$  (small) has centre  $B$  at distance  $d = 100$  m from  $A$  and centreline radius  $R_B = \rho + w = 22$  m, where the track half-width  $w = (R - r)/2 = 2$  m and  $\rho = 20$  m is the inner-wall radius at  $B$  from Figure 1 of the problem statement. The straights are inclined at  $\alpha = \arcsin((R_A - R_B)/d) = 15.0701^\circ$ , giving the segment lengths

$$\begin{aligned} l_{\text{tan}} &= \sqrt{d^2 - (R_A - R_B)^2} = 96.5609 \text{ m}, \\ l_{\text{arc},A} &= R_A(\pi + 2\alpha) = 176.0466 \text{ m}, \\ l_{\text{arc},B} &= R_B(\pi - 2\alpha) = 57.5421 \text{ m}, \end{aligned} \tag{1}$$

giving a total centreline perimeter  $L = 2l_{\text{tan}} + l_{\text{arc},A} + l_{\text{arc},B} = 426.7104$  m.

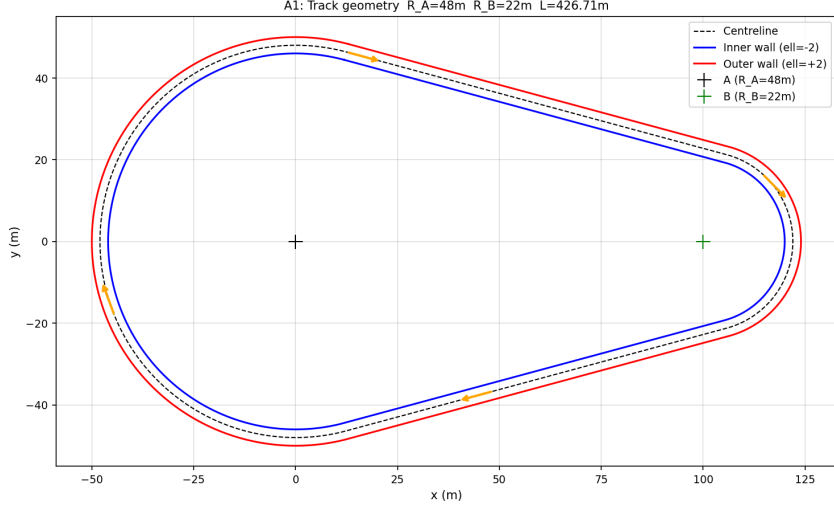
The vehicle position is parameterised by arc-length  $s \in [0, L)$  (clockwise from the upper straight) and signed lateral offset  $\ell \in [-2, +2]$  m ( $\ell > 0$ : outward); three piecewise-analytic functions  $p(s)$ ,  $n(s)$  and  $p(s, \ell) = p(s) + \ell n(s)$  implement the geometry. Five verification tests pass: the track width is 4.0000 m everywhere, unit-normal magnitudes are 1 to six decimals, the arc-straight join gaps are below the 2 mm probe resolution, the shoelace area is negative ( $-11\,617.32 \text{ m}^2$ , confirming clockwise traversal), and  $\Delta x = +4.83$  m at  $s = 0$ . The geometry is shown in Figure 1.

### 1.2 Vehicle Motion Model

The state vector is  $z = [s, \ell, v_s, v_\ell]^\top$ , where  $v_s$  and  $v_\ell$  denote the longitudinal and lateral velocities. The discrete-time nonlinear state equation is

$$z_{t+1} = f(z_t, w_t), \quad w_t \sim \mathcal{N}(0, Q), \tag{2}$$

with sampling time  $h = 0.01$  s (100 Hz, fixed by the problem statement).



**Figure 1:** Track geometry: centreline (dashed), inner wall (blue,  $\ell = -2$  m), outer wall (red,  $\ell = +2$  m), and arc centres  $A$ ,  $B$ . Orange arrows indicate clockwise travel.

**Bound-respecting nonlinear dynamics.** Rather than enforcing the physical constraints  $|\ell| < 2$  m,  $|v_\ell| \leq 0.55$  m/s,  $v_s \in (0, 25]$  m/s by post-hoc clipping, the model enforces them *by construction* through three smooth  $C^\infty$  mechanisms: (i) a position-dependent stiffness  $k(\ell) = k_2(1 + \beta(\ell^2/(\ell_{\max}^2 - \ell^2))^2)$  that diverges as  $|\ell| \rightarrow \ell_{\max} = 2$  m, modelling the track walls as elastic barriers; (ii) a tanh saturation keeping  $|v_\ell| \leq 0.55$  m/s, modelling tyre-friction and driver limits; and (iii) a tanh saturation keeping  $v_s \in (0, 25]$  m/s, modelling engine/brake capability. At the nominal operating point ( $\ell = 0$ ,  $v_\ell = 0$ ) the dynamics reduce to the familiar linear spring–damper form, so normal operation is indistinguishable from a linear Kalman filter; the nonlinear terms activate only near the physical bounds. Because all saturations are  $C^\infty$ , the dynamics remain differentiable everywhere, preserving the validity of the EKF linearisation used in Section 2.

**Linearisation at the nominal operating point.** At the nominal operating point ( $\ell = 0$ ,  $v_\ell = 0$ ,  $v_s = 10$  m/s) the effective local Jacobian of the bound-respecting dynamics evaluates to

$$A_0 = \begin{bmatrix} 1 & 0 & h & 0 \\ 0 & 1 & 0 & h \\ 0 & 0 & \phi_s \operatorname{sech}^2(\zeta_s) & 0 \\ 0 & -hk_2 & 0 & \phi_1 \end{bmatrix}, \quad G = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad (3)$$

where the (3, 3) entry is the pre-saturation autoregressive coefficient  $\phi_s = 0.999$  multiplied by the local slope of the smooth tanh speed-bounding map (so  $\phi_s \operatorname{sech}^2(\zeta_s) \approx 0.960$  at  $v_s = 10$  m/s, with  $\zeta_s$  the argument of the tanh saturation evaluated at the nominal speed),  $\phi_1 = 0.9$  and  $k_2 = 10$ . Process noise enters only the two velocity channels, so  $Q \in \mathbb{R}^{2 \times 2}$  and  $G \in \mathbb{R}^{4 \times 2}$ ; the  $s$  and  $\ell$  states are propagated through the deterministic kinematic integrators. The four eigenvalues  $\{0.9113, 0.9887, 1, 0.9601\}$  all satisfy  $|\lambda| \leq 1$ ; the unit eigenvalue is the open-loop arc-length integrator and is not strictly stable, so the Lyapunov cross-check below is applied only to the strictly stable  $[\ell, v_s, v_\ell]$  sub-system. The matrix  $G$  is the *unsaturated* reference noise-injection matrix; the EKF time update of Section 2.1 uses the local noise-input Jacobian  $L_t = \partial f / \partial w|_{\hat{z}_t|_t}$  instead, whose third diagonal entry carries the same  $\operatorname{sech}^2$  saturation slope as  $A_t$ .  $A_0$  and  $G$  are used here only for the Lyapunov steady-state cross-check of Section 1.5.

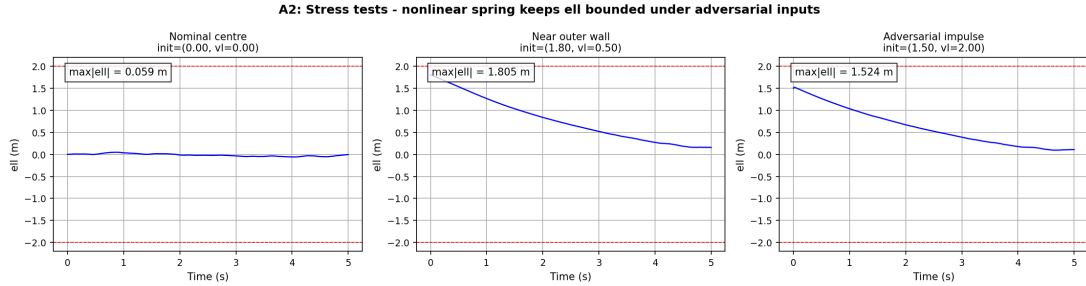
**Parameters.** Process noise  $Q = \operatorname{diag}(0.5, 0.002)$ , justified by the Monte-Carlo scans of Section 1.4; measurement noise  $R = \sigma_y^2 = 1.5^2 = 2.25$  m<sup>2</sup>; initial covariance  $P_0 = \operatorname{diag}(100, 1, 25, 0.5)$ ; initial state  $z_0 = [0, 0, 10, 0]^\top$ .

**Monte-Carlo verification.** A Monte-Carlo study (100 trials, 3000 steps each) with the full nonlinear dynamics verifies all physical constraints with substantial margin (Table 1); no post-hoc clipping is applied.

**Table 1:** Monte-Carlo constraint verification over 100 trials  $\times$  3000 steps with the bound-respecting nonlinear model. No clipping was applied: reported maxima are the true dynamics output.

| Constraint      | Bound         | Observed  | Status |
|-----------------|---------------|-----------|--------|
| $\max  \ell $   | $< 2.0$ m     | 0.109 m   | Pass   |
| $\max  v_\ell $ | $< 0.55$ m/s  | 0.335 m/s | Pass   |
| $\max v_s$      | $\leq 25$ m/s | 20.08 m/s | Pass   |
| $\min v_s$      | $> 0$ m/s     | 5.05 m/s  | Pass   |

**Adversarial stress tests.** To verify that bound enforcement is active rather than latent, the model is integrated from three initial conditions: nominal centre, near the outer wall ( $\ell_0 = 1.8$  m,  $v_{\ell,0} = 0.5$  m/s), and an adversarial impulse ( $\ell_0 = 1.5$  m,  $v_{\ell,0} = 2.0$  m/s — far outside the problem-statement bound on  $v_\ell$ ). In all three cases the trajectory remains strictly inside  $|\ell| < 2$  m (Figure 2), confirming that the nonlinear spring actively returns the vehicle to the centreline even from initial conditions beyond the linear regime.



**Figure 2:** Adversarial stress tests: lateral offset  $\ell(t)$  for three initial conditions, including an impulse  $v_{\ell,0} = 2.0$  m/s beyond the operational bound. The nonlinear spring keeps  $|\ell| < 2$  m in all cases.

### 1.3 Beacon Placement Optimisation

**GDOP criterion.** Beacon-layout quality is quantified by the Geometric Dilution of Precision [1]:

$$\text{GDOP}(p) = \sqrt{\text{tr}[(H(p)^\top H(p))^{-1}]}, \quad H_i(p) = \frac{p - a_i}{\|p - a_i\|}, \quad (4)$$

so that the expected position error equals  $\text{GDOP} \times \sigma_y$  and  $\text{GDOP} < 1$  *attenuates* sensor noise. The selection criterion is the smallest beacon count for which  $\text{GDOP}(p) < 1$  at every sampled track position.

**Connection to estimation theory.** Following the BLUE development in the course [7], for distance measurements  $y_i = \|p - a_i\| + v_i$  with  $v_i \sim \mathcal{N}(0, \sigma_y^2)$  the Fisher information matrix at position  $p$  is  $I(p) = H(p)^\top H(p) / \sigma_y^2$ , and the Cramér–Rao lower bound satisfies

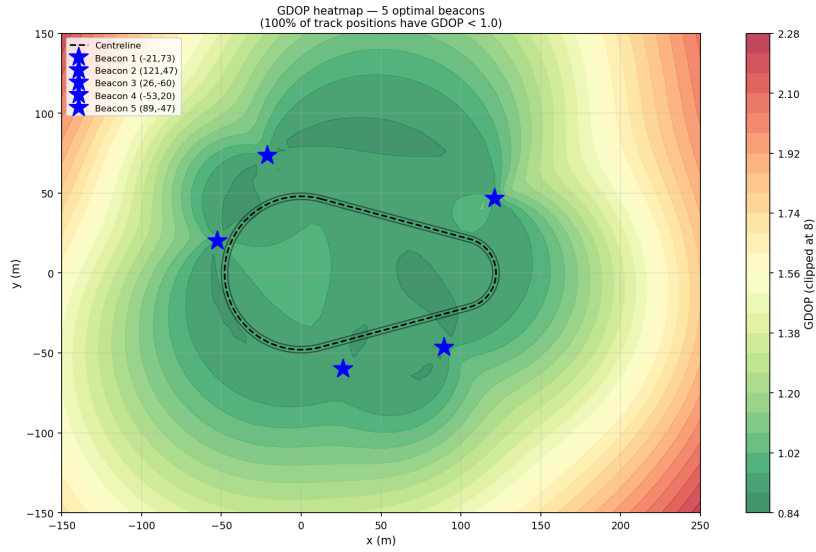
$$\text{tr}[\text{CRLB}(p)] = \sigma_y^2 \text{tr}[(H^\top H)^{-1}] = \sigma_y^2 \text{GDOP}(p)^2. \quad (5)$$

Thus GDOP is the geometry-only scalar summary of the Fisher information available from a single set of distance measurements. The quantity  $\sigma_y \cdot \text{GDOP} = 1.379$  m is used in Section 2.4 as a single-measurement GDOP benchmark, that is, a geometry-derived reference for isolated range readings rather than a lower bound on the recursive EKF, which integrates information across many time steps through the motion model.

**Algorithm and beacon count.** Candidates are drawn from a  $20 \times 16$  grid restricted to  $[w + 5, w + 30]$  m from the track boundary; the outer limit of 30 m is the Pareto-optimal knee identified by the spatial ablation of Figure 4. A greedy algorithm with exhaustive two-beacon initialisation and sequential single-beacon additions is used as a tractable heuristic motivated by standard submodular sensor-selection methods [3]; submodularity of the GDOP objective is not proved here, so the layout is validated empirically by a swap refinement pass (which yields only a 1.4% improvement, indicating proximity to a local optimum), a beacon-count ablation, a single-beacon failure analysis, and an installation-distance sensitivity study. Table 2 shows an abrupt threshold at  $N = 5$ : with  $N = 4$ , 0% of track positions satisfy  $\text{GDOP} < 1$ ; with  $N = 5$ , 100% do. The five selected coordinates (in m) are  $(-21.1, 73.3)$ ,  $(121.1, 46.7)$ ,  $(26.3, -60.0)$ ,  $(-52.6, 20.0)$ ,  $(89.5, -46.7)$ ; see Figure 3.

**Table 2:** GDOP statistics by beacon count ( $\sigma_y = 1.5$  m).

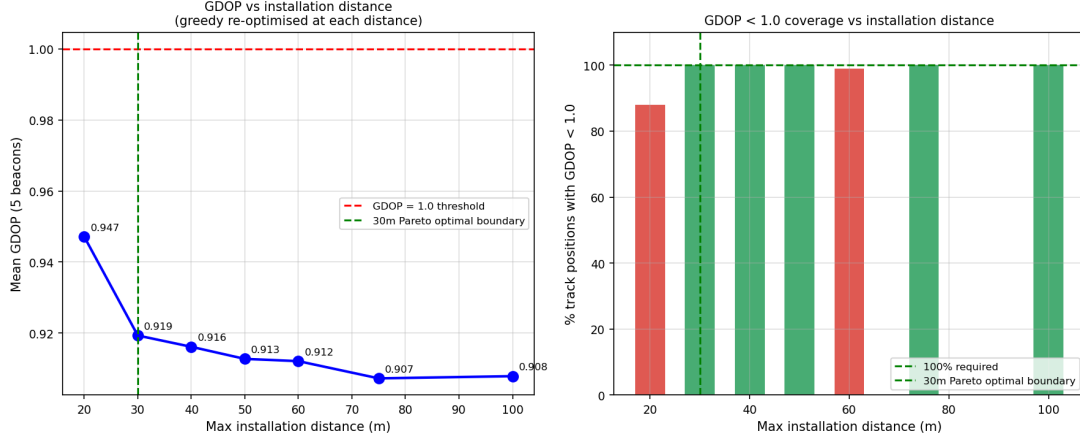
| $N$ | Mean GDOP | Max GDOP | Expected error | % with $\text{GDOP} < 1$ |
|-----|-----------|----------|----------------|--------------------------|
| 2   | 1.7177    | 2.4805   | 2.577 m        | —                        |
| 3   | 1.2108    | 1.3997   | 1.816 m        | —                        |
| 4   | 1.0475    | 1.1885   | 1.571 m        | 0%                       |
| 5   | 0.9193    | 0.9770   | 1.379 m        | 100%                     |
| 6   | 0.8292    | 0.8951   | 1.244 m        | 100%                     |



**Figure 3:** GDOP heatmap for the five optimal beacons (blue stars). The track lies entirely within the attenuating zone ( $\text{GDOP} < 1$ , green); warm colours outside the track are not part of the objective.

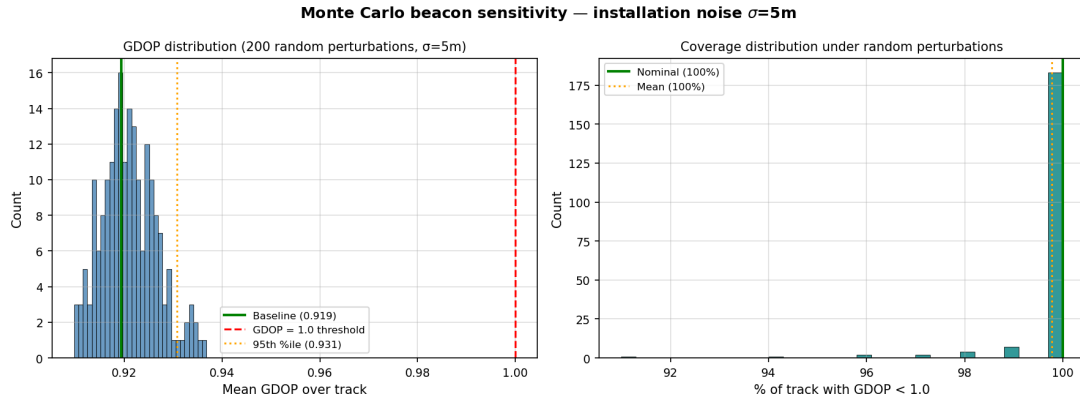
**Single-beacon failure robustness.** Removing any one of the five beacons raises the mean GDOP from 0.9193 to between 1.048 and 1.087 (+13.9–18.3%, worst case  $a_1$ ); coverage drops from 100% to 0% but mean GDOP remains below 1.09. The  $N = 5$  configuration is therefore selected because it nominally meets the  $\text{GDOP} < 1$  criterion and fails gracefully under any single-point fault.

**Installation boundary (spatial ablation).** The 30 m outer limit is data-derived (Figure 4). At 20 m, only 88% of the track achieves  $\text{GDOP} < 1$ ; at 30 m coverage reaches 100% with the largest single gain (+2.9%); beyond 30 m improvement falls below 0.5%. Minor non-monotonicity at 60 m and 100 m is a greedy-search artefact (different initial pairs are selected as the candidate pool grows). The 30 m boundary is therefore Pareto-optimal: the smallest outer limit that meets the GDOP criterion everywhere.



**Figure 4:** Spatial ablation: mean GDOP (left) and the fraction of track with GDOP < 1 (right) versus maximum installation distance, with the greedy algorithm re-optimised at each boundary. The dashed line at 30 m marks the Pareto-optimal knee where 100% coverage is first achieved.

**Monte-Carlo sensitivity to installation error.** Installation errors are stochastic in practice (GPS accuracy, surveyor tolerance, post-construction displacement). A Monte-Carlo study with 200 realisations, each perturbing every beacon coordinate independently by  $\mathcal{N}(0, 5^2)$  m, yields median mean-GDOP 0.9205 (+0.1%), 95th percentile 0.9308 (+1.2%), and  $P(\text{GDOP} < 1) = 100\%$  across all realisations (Figure 5). The design is therefore robust to realistic installation noise.



**Figure 5:** Monte-Carlo beacon sensitivity under independent  $\mathcal{N}(0, 5^2)$  installation errors per coordinate (200 realisations). All realisations satisfy GDOP < 1.

## 1.4 Noise Parameter Validation

Three Monte-Carlo studies (10 trials for the  $Q$  and  $(\phi_1, k_2)$  scans, 20 trials for the  $h$  ablation, each  $\times 3000$  steps, common random numbers for variance reduction) validate the parameter choices of Section 1.2 (Table 3). The selected operating point lies in the interior of a wide safe basin in each scan, with all three constraints satisfied by substantial margin. Full heatmaps for the  $(q_s, q_\ell)$  and  $(\phi_1, k_2)$  scans, together with the sampling-time ablation, are provided in the GitHub repository.

The tuning reflects the standard Kalman trade-off:  $Q \rightarrow 0$  drives the Kalman gain to zero and causes the filter to ignore measurements, while excessively large  $Q$  destroys smoothing. The  $h$  ablation confirms that the bound-respecting model is robust across the range  $h \in [0.001, 0.1]$  s; the  $h = 0.01$  s choice follows the problem statement, and the model does not demand it uniquely.

**Table 3:** Parameter validation summary at the selected operating point. Bounds:  $|\ell| < 2.0$  m,  $|v_\ell| < 0.55$  m/s.

| Validation study                          | Max $ \ell $ | Max $ v_\ell $ | Status |
|-------------------------------------------|--------------|----------------|--------|
| $Q$ scan at $Q = \text{diag}(0.5, 0.002)$ | 0.096 m      | 0.295 m/s      | Pass   |
| $(\phi_1, k_2)$ scan at $(0.9, 10)$       | 0.096 m      | 0.295 m/s      | Pass   |
| $h$ ablation at $h = 0.01$ s (100 Hz)     | 0.109 m      | 0.314 m/s      | Pass   |

## 1.5 Sensor Bias Handling via State Augmentation

Each beacon may exhibit a constant additive measurement bias  $\beta_i$  due to hardware calibration errors or multipath effects. If ignored, the EKF estimate drifts persistently. Following the state-augmentation approach of [8, 10] (adapted from dynamics bias to measurement bias) and [2] (Ch. 13), the state is augmented to  $z_{\text{aug}} = [s, \ell, v_s, v_\ell, \beta_1, \dots, \beta_5]^\top \in \mathbb{R}^9$ , with dynamics  $z_{\text{aug},t+1} = A_{\text{aug}} z_{\text{aug},t} + G_{\text{aug}} w_{\text{aug},t}$  and

$$\begin{aligned} A_{\text{aug}} &= \begin{bmatrix} A & 0 \\ 0 & I_5 \end{bmatrix}, \quad G_{\text{aug}} = \begin{bmatrix} G & 0 \\ 0 & I_5 \end{bmatrix}, \\ Q_{\text{aug}} &= \text{diag}(0.5, 0.002, \underbrace{10^{-6}, \dots, 10^{-6}}_5). \end{aligned} \quad (6)$$

The  $I_5$  block in  $A_{\text{aug}}$  models each bias as a random walk. The  $I_5$  block in  $G_{\text{aug}}$  assigns each beacon an *independent* noise channel: the five hardware units have uncorrelated internal drifts, and a shared channel would incorrectly imply 100% correlation. Initial bias uncertainty is  $P_{0,\beta_i} = 25 \text{ m}^2$  ( $\sigma_\beta = 5$  m). The augmented measurement function is  $h_i(z_{\text{aug}}) = \|p(s, \ell) - a_i\| + \beta_i$ , with bias Jacobian  $\partial h_i / \partial \beta_j = \delta_{ij}$  (identity block  $I_5$ ). Numerical verification with known biases recovered each bias to within  $6 \times 10^{-15}$  m.

**Static observability and practical bias estimation.** The frozen-linearisation observability matrix  $\mathcal{O} = [C; CA_{\text{aug}}; \dots; CA_{\text{aug}}^8]$  satisfies  $\text{rank}(\mathcal{O}) = 8$ , one short of the state dimension. Two structural causes contribute:  $C$  has zero columns for  $v_s$  and  $v_\ell$  (range does not depend on velocity directly), and the dynamic coupling  $CA_{\text{aug}}$  adds only one further independent direction through the position–velocity integrator. This static test therefore does *not* prove full joint observability of the vehicle state and all five biases at any single instant, and the augmented filter is not claimed to be globally observable from a single linearisation. The rank-8 result is, however, a frozen-linearisation diagnostic, not a statement about the moving nonlinear system over a complete lap: along a trajectory the measurement Jacobian  $C_t$  changes as the vehicle moves relative to the five beacons, and information accumulates through the time-varying observability Gramian involving  $C_t \Phi_{t,k}$ . A direction that is weakly observable in any single frozen geometry can therefore become identifiable over the full trajectory; the recovery of all five injected biases under noise in Section 4 provides this trajectory-level empirical verification [2].

**Steady-state cross-check.** Following [6], for the strictly stable  $[\ell, v_s, v_\ell]$  sub-system the covariance converges to the unique stationary value  $P_\infty$  satisfying the discrete-time Lyapunov equation  $P_\infty = A_{\text{sub}} P_\infty A_{\text{sub}}^\top + G_{\text{sub}} Q G_{\text{sub}}^\top$ , giving  $\sigma_{\ell,\infty} = 0.032$  m,  $\sigma_{v_\ell,\infty} = 0.103$  m/s, and a mean-reversion standard deviation  $\sigma_{v_s,\infty} = 2.53$  m/s about the target  $v_s = 10$  m/s. The  $3\sigma$  envelopes (0.095 m and 0.309 m/s) agree with the nonlinear Monte-Carlo maxima (0.109 m and 0.335 m/s) to within the expected tail factor, confirming that  $A(\hat{z})$  is a faithful local linearisation at the nominal operating point — which is precisely what the EKF relies on at every prediction step. The tanh saturation provides additional protection against off-nominal excursions that the linear model cannot rule out a priori (Figure 2).

## 2 Extended Kalman Filter

The estimator chapters that follow build directly on the modelling component of Section 1: the linearised dynamics matrices  $A_0, G$  at the nominal operating point and the process/measurement noise covariances

$Q = \text{diag}(0.5, 0.002)$  and  $R = 2.25 \text{ m}^2 \cdot I_5$  (Section 1.2); the five GDOP-optimised beacon coordinates and the resulting Cramér–Rao lower bound  $\sigma_y \cdot \overline{\text{GDOP}} = 1.379 \text{ m}$  on single-shot position RMSE (Section 1.3); the Lyapunov steady-state standard deviations  $\sigma_{\ell, \infty} = 0.032 \text{ m}$ ,  $\sigma_{v_{\ell}, \infty} = 0.103 \text{ m/s}$ ,  $\sigma_{v_s, \infty} = 2.53 \text{ m/s}$  used here as theoretical references (Section 1.5); the augmented-state formulation  $z_{\text{aug}} = [s, \ell, v_s, v_{\ell}, \beta_1, \dots, \beta_5]^\top$  with its noise-free  $h_{\text{aug}}$  verification (Section 1.5, implemented numerically in Section 4); and the full nonlinear truth model  $f(\cdot)$  of Section 1.2, used as the simulator throughout all estimator validation experiments in this chapter.

## 2.1 EKF Formulation

Both the state transition  $f(\cdot)$  from Section 1.2 and the range-measurement function  $h(\cdot)$  are genuinely nonlinear:  $f$  contains the position-dependent stiffness  $k(\ell)$  and the smooth tanh saturations on  $v_s$  and  $v_{\ell}$ , while  $h$  returns the Euclidean distance from the vehicle to each of the five anchors through the piecewise track geometry of Section 1.1. The standard Kalman filter is therefore insufficient; an EKF that linearises both  $f$  and  $h$  at the current estimate is required, in exact correspondence with the nonlinear branch of the lecturer’s notebook [11].

Following the convention of [11], the measurement update executed when a measurement  $y_t$  is received is

$$\nu_t = y_t - h(\hat{z}_{t|t-1}), \quad (7)$$

$$S_t = C_t \Sigma_{t|t-1} C_t^\top + R, \quad (8)$$

$$\hat{z}_{t|t} = \hat{z}_{t|t-1} + \Sigma_{t|t-1} C_t^\top S_t^{-1} \nu_t, \quad (9)$$

$$\Sigma_{t|t} = \Sigma_{t|t-1} - \Sigma_{t|t-1} C_t^\top S_t^{-1} C_t \Sigma_{t|t-1}, \quad (10)$$

where  $C_t = \partial h / \partial z|_{\hat{z}_{t|t-1}}$  is the measurement Jacobian and the additive-noise structure makes  $J_v h \equiv I_5$ , so the linearised innovation covariance reduces to  $C_t \Sigma_{t|t-1} C_t^\top + R$  with no auxiliary rescaling. The time update propagates the estimate forward through the *full* nonlinear dynamics

$$\hat{z}_{t+1|t} = f(\hat{z}_{t|t}, 0), \quad (11)$$

$$\Sigma_{t+1|t} = A_t \Sigma_{t|t} A_t^\top + L_t Q L_t^\top, \quad (12)$$

with the prediction-step Jacobians  $A_t = \partial f / \partial z|_{\hat{z}_{t|t}}$  and  $L_t = \partial f / \partial w|_{\hat{z}_{t|t}}$  both evaluated at the current state estimate at every time step. The state-input Jacobian  $A_t$  accounts for the local effect of the nonlinear stiffness  $k(\ell)$  when the vehicle is near a boundary; the noise-input Jacobian  $L_t$  propagates process-noise covariance through the same tanh-saturated channels that the truth model itself uses, so the predictor and the simulator have identical noise gains by construction. The two Jacobians coincide with the closed-form  $A_0$  and  $G$  of Section 1.2 only at the nominal operating point; everywhere else they differ, and using the local Jacobians is what removes the predictor–simulator mismatch that an  $A_0$ -frozen prediction would carry. All four Jacobians ( $\partial h / \partial z$ ,  $\partial h / \partial v$ ,  $\partial f / \partial z$ ,  $\partial f / \partial w$ ) are obtained from CasADi automatic differentiation [4]; the measurement function is built piecewise-symbolically via `cs.if_else` so that the four track segments (two straights and two arcs) of Section 1.1 are handled exactly across segment transitions, with no finite-difference discretisation error. The variable names (`x_t_pred`, `sigma_t_pred`, `x_t_meas`, `sigma_t_meas`) and numbered loop comments follow the lecturer’s convention so that the implementation is auditable against the lecture material. The setup layer of `Estimator.ipynb` configures the track geometry, anchor coordinates, the noise covariances  $Q$  and  $R$ , the CasADi symbolic functions  $h$ ,  $\dot{h}_x$ ,  $\dot{h}_v$ ,  $\dot{f}_x$ ,  $\dot{f}_w$ ,  $f_{\text{fun}}$ , and the EKF primitives `measurement_update` and `time_update`. All quantities are imported from the modelling component (Section 1); the layer is reused by every subsequent cell without modification.

## 2.2 Internal Model Alignment

The longitudinal velocity dynamics specified by the simulator are autoregressive of the first order, centred at the nominal speed  $v_{s,\text{tgt}} = 10 \text{ m/s}$  rather than zero,

$$v_{s,t+1} = \phi_s v_{s,t} + (1 - \phi_s) v_{s,\text{tgt}} + w_{s,t}. \quad (13)$$

The constant term  $(1 - \phi_s) v_{s,\text{tgt}}$  in (13) would not appear in any *linear* time-update that uses only the state-transition matrix  $A_0$ . An early development iteration that began from such a linear  $A_0$ -only predictor exhibited a small but persistent bias in the longitudinal velocity estimate; inspection of the mean velocity trajectories produced by the simulator versus the predictor isolated the missing affine term as the cause. The current implementation removes the issue by construction: because the predictor uses the full nonlinear function  $\hat{z}_{t+1|t} = f(\hat{z}_{t|t}, 0)$  from (11) rather than  $A_0 \hat{z}_{t|t}$ , the affine mean-reversion term in (13) is part of  $f$  itself and propagates without any explicit correction. The same logic applies to the lateral restoring force  $-h k(\ell)\ell$  and the tanh saturations: all are contained inside the single function call  $f(\cdot, 0)$  that the predictor evaluates, so the EKF’s internal model is by definition identical to the simulator’s truth dynamics. The NIS consistency check in Section 2.5 confirms statistically that no residual internal-model bias remains.

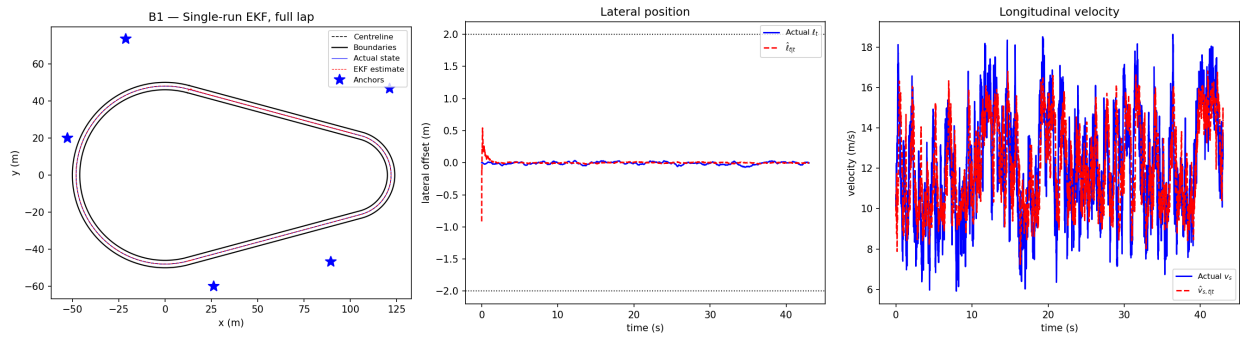
### 2.3 Single-Run Simulation

A representative single-lap simulation was conducted over  $N = 4300$  time steps (approximately 43 s at the nominal speed). The initial state is  $z_0 = [0, 0, 10, 0]^\top$  with deliberately diffuse initial covariance

$$P_0 = \text{diag}(100, 1, 25, 0.5) = \text{diag}(10^2, 1^2, 5^2, 0.707^2), \quad (14)$$

which corresponds to standard-deviation priors of approximately 10 m on arc-length (about 1 s of travel at the nominal 10 m/s), 1 m on lateral offset (one half-lane-width), 5 m/s on longitudinal velocity (50% of the nominal speed), and 0.707 m/s on lateral velocity (slightly above the 0.55 m/s physical bound). The diffuseness is deliberate: the recursive filter forgets its prior within several hundred steps, as the closed-loop covariance trace of Figure 7 (Section 2.3) and a five-point  $P_0$ -scaling scan (steady-state arc error invariant to within  $10^{-7}$  m across  $0.1 \times -5 \times P_0$ ) both confirm.

The remaining design parameters use “hard” problem-statement quantities wherever possible:  $h = 1/100 = 0.01$  s is fixed by the 100 Hz beacon update rate, and  $R = \sigma_y^2 I_5 = 1.5^2 I_5 = 2.25 I_5 \text{ m}^2$  uses the problem-statement range standard deviation  $\sigma_y = 1.5$  m. The process-noise covariance  $Q = \text{diag}(0.5, 0.002)$  is not specified by the problem and was selected by the modelling-side parameter scans of Section 1.4; the NIS consistency check in Section 2.5 below provides an independent estimator-side confirmation that this  $Q$  matches the true noise statistics to within 0.1% of the theoretical NIS mean. Results are shown in Figure 6.

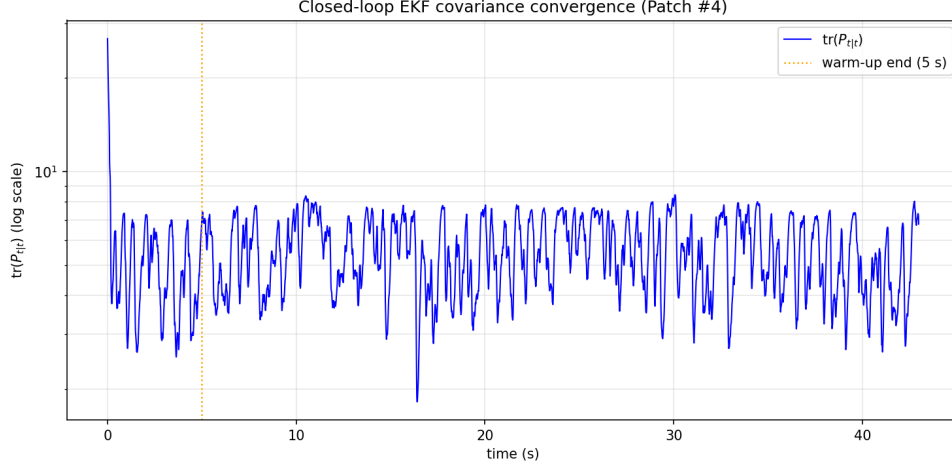


**Figure 6:** Single-run EKF over one full lap. Left: estimated trajectory (red dashed) overlaid on truth (blue) with anchors as stars. Centre: lateral-position tracking with the  $\pm 2$  m bound. Right: longitudinal-velocity tracking.

The estimated trajectory closely follows the true path throughout both straight sections and the asymmetric arcs, and the lateral estimate remains within the  $\pm 2$  m boundary. A brief transient is visible in the first 0.5 s as the covariance converges from the diffuse prior. The mean longitudinal velocity error of 2.024 m/s is larger in absolute terms than the position errors; this reflects both the Kalman filter’s one-step tracking delay under a process-noise variance of  $q_s = 0.5$  and, more fundamentally, the Lyapunov bound established in Section 1.5:  $\sigma_{v_s, \infty} = 2.53$  m/s. The observed value is approximately 80% of this bound, consistent with a near-optimal filter. The arc-length error of 0.239 m over a track perimeter of 426.7 m represents a relative accuracy of 0.056%; residual velocity error does not propagate into systematic position bias.



**Closed-loop covariance convergence.** The trace of the recursive posterior covariance  $\text{tr}(P_{t|t})$  contracts from its diffuse-prior initial value of approximately 26.8 to a steady-state mean of 5.55 over the warm-up window (Figure 7), a  $\sim 5\times$  decay that empirically verifies closed-loop convergence of the EKF without invoking the linearisation analysis of Section 1.5. After the warm-up the trace is essentially constant, consistent with the existence of the Lyapunov stationary covariance derived in Section 1.5.



**Figure 7:** Closed-loop EKF posterior covariance trace  $\text{tr}(P_{t|t})$  over the single-lap run, log-scaled. The trace decays by approximately  $5\times$  during the 5 s warm-up window (orange dotted line) and is stationary thereafter.

**Sensitivity to the initial covariance  $P_0$ .** The diffuse-prior values  $P_0 = \text{diag}(100, 1, 25, 0.5)$  are an intentional choice rather than a tuned one. Scaling  $P_0$  jointly by factors in  $\{0.1, 0.5, 1, 2, 5\}$  leaves the warm-up-excluded mean arc-length error invariant to within  $10^{-7}$  m: the recursive filter forgets its initial covariance after a few hundred steps regardless of how diffuse or tight the prior is, so the choice affects only the warm-up duration and not steady-state accuracy.

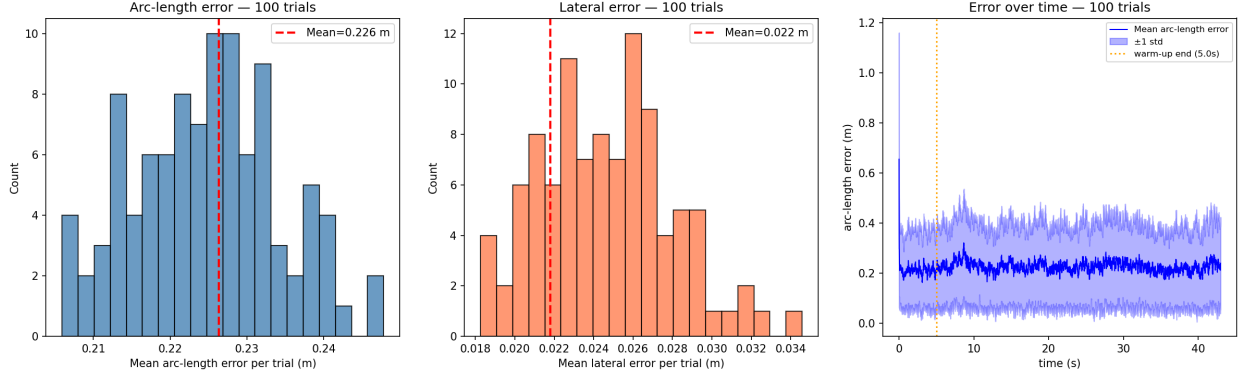
## 2.4 Monte-Carlo Statistical Validation

To verify that the observed performance is not the result of a single favourable noise realisation, a Monte-Carlo study was conducted with  $N_{\text{MC}} = 100$  independent trials, each one full lap in length. The trial count was chosen to keep the standard error of the reported mean small relative to the mean itself: with the across-trial standard deviation of arc-length error  $\sigma_{\text{trial}} = 0.010$  m, the standard error of the mean is  $\sigma_{\text{trial}}/\sqrt{N_{\text{MC}}} = 0.001$  m and the 95% confidence half-width is  $\approx 0.002$  m, two orders of magnitude below the mean of 0.226 m. The first 5 s (500 time steps) of each trial were excluded from the statistics to separate steady-state performance from the covariance-convergence transient; the cutoff is empirically calibrated against the single-run covariance trace of Figure 7, which decays from  $\text{tr}(P_{0|0}) \approx 26.8$  to a steady-state mean of 5.55 within this window. Results are summarised in Table 4 and Figure 8.

**Table 4:** EKF performance statistics over 100 Monte-Carlo trials. Standard deviation is computed across per-trial means. Statistics exclude the first 5 s of each trial (warm-up).

| Metric               | Mean  | Std (across trials) | Max   |
|----------------------|-------|---------------------|-------|
| Arc-length error (m) | 0.226 | 0.010               | 1.265 |
| Lateral error (m)    | 0.022 | 0.003               | 0.103 |
| Velocity error (m/s) | 1.753 | 0.072               | —     |

Two independent comparisons against the modelling-side references support near-optimal operation. The first is a scale check rather than a violated bound: the recursive EKF arc-length error of 0.226 m sits well below the  $\sigma_y \cdot \overline{\text{GDOP}} = 1.379$  m single-measurement GDOP scale of Section 1.3. This comparison is only



**Figure 8:** Monte-Carlo validation over 100 trials: per-trial mean arc-length and lateral error histograms (left, centre) and the trial-averaged arc error vs. time with  $\pm 1$  std band (right).

a reference:  $\sigma_y \overline{\text{GDOP}}$  is a two-dimensional, single-shot position benchmark, whereas the reported EKF statistic is a recursive along-track error. The smaller EKF value is attributed jointly to temporal filtering and to the one-dimensional projection, not to a violation of the GDOP / Fisher-information benchmark. The second comparison is tighter: the Lyapunov analysis of Section 1.5 gives references  $\sigma_{\ell, \infty} = 0.032$  m and  $\sigma_{v_{\ell}, \infty} = 0.103$  m/s for the strictly stable lateral subsystem, and the EKF’s empirical mean lateral error of 0.022 m is comparable to and slightly below  $\sigma_{\ell, \infty}$ , with the deviation within the 0.003 m across-trial standard deviation. Together the two comparisons indicate that the recursive filter operates close to the modelling-side covariance reference for lateral accuracy.

## 2.5 NIS Consistency Check

Performance metrics such as RMSE confirm that the filter is accurate *in this run*, but they do not directly verify that the design covariances  $Q$  and  $R$  match the true noise statistics. A filter could achieve a low RMSE while being internally miscalibrated — over- or under-confident in its own uncertainty — and the miscalibration would only surface under stress. The standard diagnostic for filter consistency [12] is the Normalized Innovation Squared (NIS) test:

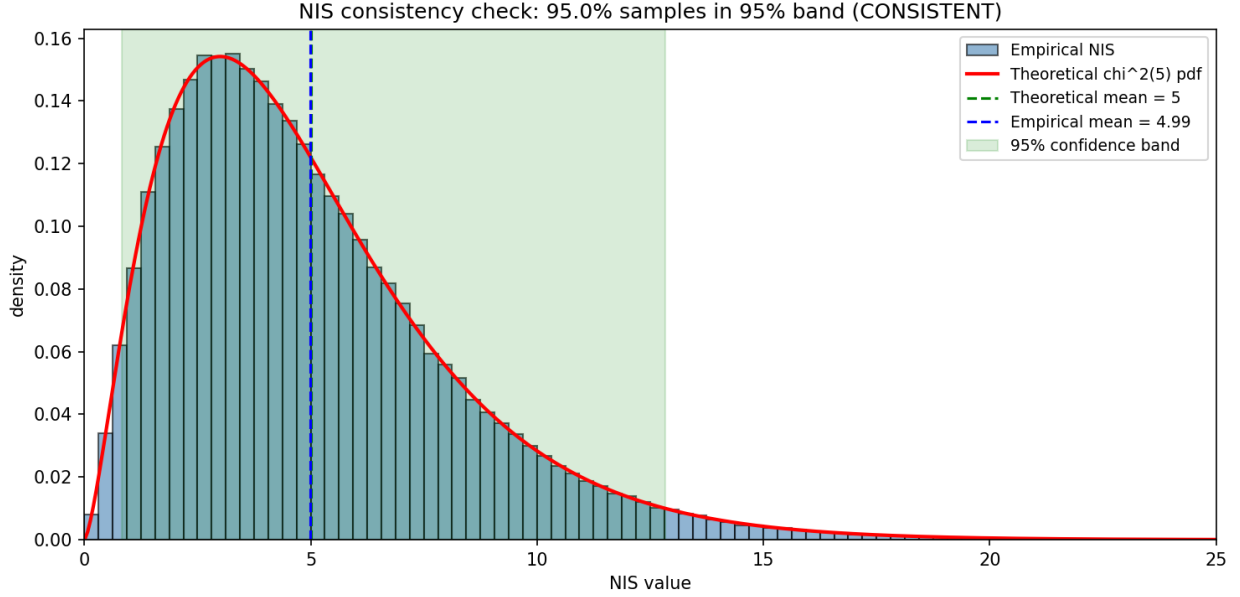
$$\begin{aligned} \nu_t &= y_t - h(\hat{z}_{t|t-1}), \\ S_t &= C_t \Sigma_{t|t-1} C_t^\top + R, \\ \text{NIS}_t &= \nu_t^\top S_t^{-1} \nu_t. \end{aligned} \tag{15}$$

If the filter is correctly tuned,  $\text{NIS}_t$  follows a chi-square distribution with  $n_y = 5$  degrees of freedom (the measurement dimension), giving expected mean  $\mathbb{E}[\text{NIS}] = 5$  and a 95% confidence interval of  $[0.83, 12.83]$ .

This estimator-side consistency check is independent of and complementary to the simulator-side scan of Section 1.4: the two analyses approach the same  $Q$ -matrix selection from opposite directions, and agreement between them (shown below) provides stronger evidence than either could on its own.

NIS samples were collected from all 100 Monte-Carlo trials (excluding the warm-up window), giving approximately 380,000 samples in total. The empirical distribution is compared against the theoretical chi-square pdf in Figure 9.

The agreement is close to the theoretical value: empirical mean 4.994 versus theoretical 5.00 (relative error 0.1%), and 95.0% of samples within the 95% confidence band. The histogram shape is visually indistinguishable from the chi-square pdf across the entire range. The interpretation is that the filter is statistically consistent: the design covariances  $Q$  and  $R$  match the true noise statistics imposed by the simulator. Crucially, this provides the desired estimator-side independent confirmation of the noise-parameter selection: two independent analyses (simulator-side constraint preservation in Section 1.4 and the present estimator-side innovation statistics) both select  $Q = \text{diag}(0.5, 0.002)$ , strengthening confidence in the result.



**Figure 9:** NIS consistency check across the 100-trial Monte-Carlo dataset: empirical histogram against the theoretical  $\chi^2(5)$  pdf. Empirical mean 4.994 vs. theoretical 5.00; 95.0% of samples fall within the 95% confidence band [0.83, 12.83].

## 2.6 Beacon Count Ablation

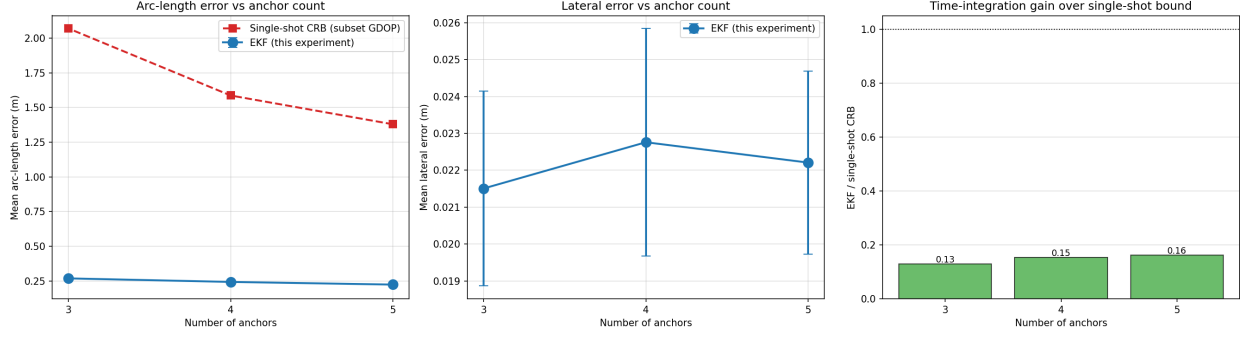
Section 1.3 selects  $N = 5$  anchors through a GDOP-based geometric analysis. To validate this choice from the estimator side, this section measures recursive EKF performance under three reduced configurations and compares each against a *self-consistent* single-shot GDOP-scale reference: rather than reusing the modelling-side optimal-greedy GDOP-scale values of Table 2 (which correspond to optimal  $N = 4$  and  $N = 3$  layouts that differ from the ones used here), the GDOP scale  $\sigma_y \overline{\text{GDOP}}$  is recomputed for the exact anchor subset retained in each configuration. This ensures Table 5 reports apples-to-apples comparisons between “EKF arc” and “GDOP scale” columns in every row.

For each of three configurations —  $N = 5$  (baseline, all anchors),  $N = 4$  (anchor  $a_3$  removed from the  $N = 5$  baseline),  $N = 3$  (anchors  $a_3$  and  $a_5$  removed) — 30 Monte-Carlo trials of one full lap were run with independent random seeds. The trial count is smaller than the 100 used in Section 2.4 because this experiment is a secondary trend / robustness check: the relevant deliverable is the monotonic ranking of the EKF error under anchor removal, not a sub-millimetre estimate of the final performance mean. Steady-state mean arc-length and lateral errors are reported in Table 5 alongside the per-subset GDOP-scale values and the EKF/GDOP-scale ratio; results are visualised in Figure 10.

**Table 5:** Beacon-count ablation: 30-trial Monte-Carlo per configuration. The GDOP-scale column is the per-subset  $\sigma_y \cdot \overline{\text{GDOP}}$  recomputed for the exact anchor subset in each row. It is used as a single-measurement geometric reference, not as a lower bound on the recursive arc-length error.

| Configuration              | EKF arc (m) | EKF lat (m) | GDOP scale (m) | EKF/GDOP scale |
|----------------------------|-------------|-------------|----------------|----------------|
| $N = 5$ (baseline)         | 0.223       | 0.0222      | 1.379          | 0.16           |
| $N = 4$ (drop $a_3$ )      | 0.243       | 0.0228      | 1.586          | 0.15           |
| $N = 3$ (drop $a_3, a_5$ ) | 0.268       | 0.0215      | 2.070          | 0.13           |

The arc-length error degrades monotonically as the anchor count is reduced ( $0.223 \rightarrow 0.243 \rightarrow 0.268$  m), in qualitative agreement with the per-subset GDOP-scale trend ( $1.379 \rightarrow 1.586 \rightarrow 2.070$  m); the EKF/GDOP-scale ratio remains in 0.13–0.16 across all three configurations, indicating that the temporal-information gain over single-shot trilateration is roughly constant. The lateral error, by contrast, is largely insensitive to



**Figure 10:** Beacon-count ablation. Left: EKF mean arc-length error (blue,  $\pm 1$  std) versus the per-subset GDOP scale (red). Centre: lateral error, largely insensitive to the count. Right: EKF/GDOP-scale ratio.

the anchor count (0.0215–0.0228 m), reflecting the fact that lateral observability depends on whether the surviving subset spans both sides of the centreline rather than on the count itself; this is a feature single-number GDOP averages do not surface but the recursive simulation does. The case for retaining  $N = 5$  is therefore not absolute accuracy — even  $N = 3$  remains well below 1 m — but robustness:  $N = 5$  degrades gracefully to  $N = 4$  under single-beacon failure, whereas  $N = 3$  degrading to  $N = 2$  produces a rank-deficient measurement Jacobian and renders the estimator infeasible. The single-beacon failure analysis of Section 1.3 and the present count ablation therefore agree on the design recommendation  $N = 5$ .

## 2.7 Near-Boundary Stress Test

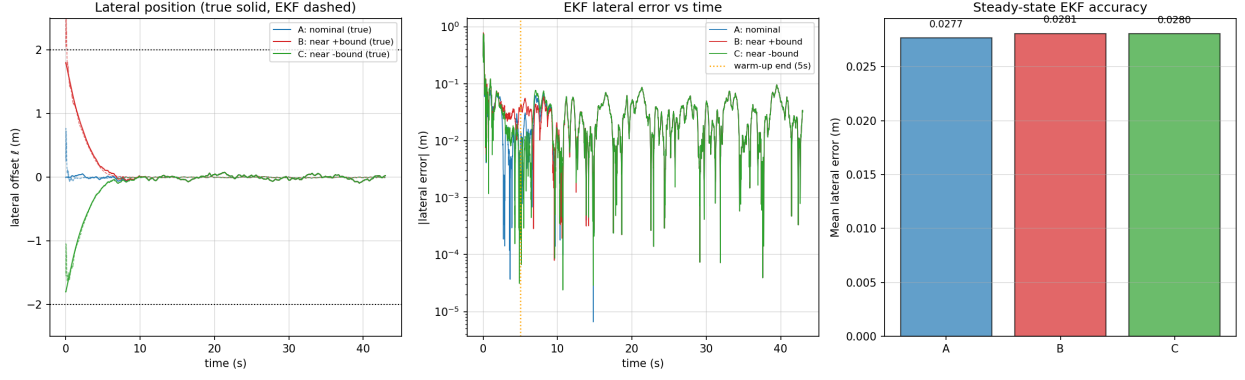
Section 1.2 includes an adversarial stress test at the simulator level: the bound-respecting nonlinear dynamics is integrated from three initial conditions to verify that the simulator’s nonlinear spring keeps  $|\ell| < 2$  m in all cases. That test answers the question “*does the simulated vehicle stay on the track?*” It does not answer the complementary question relevant for the estimator: “*does the EKF still track the vehicle accurately when the vehicle operates near the boundary?*” The latter matters because the position-dependent stiffness  $k(\ell)$  diverges as  $|\ell| \rightarrow 2$  m, so the EKF’s local linearisation is most strongly tested in this regime: the prediction-step Jacobian  $A_t = \partial f / \partial z|_{\hat{z}_t|_t}$  changes rapidly as  $|\ell|$  approaches the boundary.

The present section provides the estimator-side counterpart to the simulator-side stress test, completing a closed-loop validation: Section 1.2 confirms the dynamics keeps the vehicle on track; this section confirms the EKF can find the vehicle there. The truth model used in this experiment is the full nonlinear  $f(\cdot)$  of Section 1.2 (smooth  $C^\infty$  saturations, no post-hoc clipping), guaranteeing consistency between the modelling component and the simulator code used here.

The conditions used here differ from the adversarial impulse of Section 1.2 ( $\ell_0 = 1.5$ ,  $v_{\ell,0} = 2.0$  m/s) because the two tests have different goals. The modelling-side test stresses the dynamics with a  $v_\ell$  nearly four times the physical 0.55 m/s bound, to verify that the nonlinear spring activates in unrealistic conditions. The present test stresses the EKF under realistic but extreme operating conditions: vehicle initialised at  $\ell_0 = \pm 1.8$  m (within 0.2 m of the boundary) with  $v_{\ell,0} = \pm 0.3$  m/s (within the operational bound). Three scenarios are run, each one full lap with identical seed: A nominal ( $\ell_0 = 0$ ,  $v_{\ell,0} = 0$ ); B near +bound; C symmetric near –bound. Results are in Table 6 and Figure 11.

**Table 6:** Near-boundary stress test under the nonlinear truth model  $f(\cdot)$ : steady-state EKF lateral accuracy and maximum lateral excursion under three initial conditions.

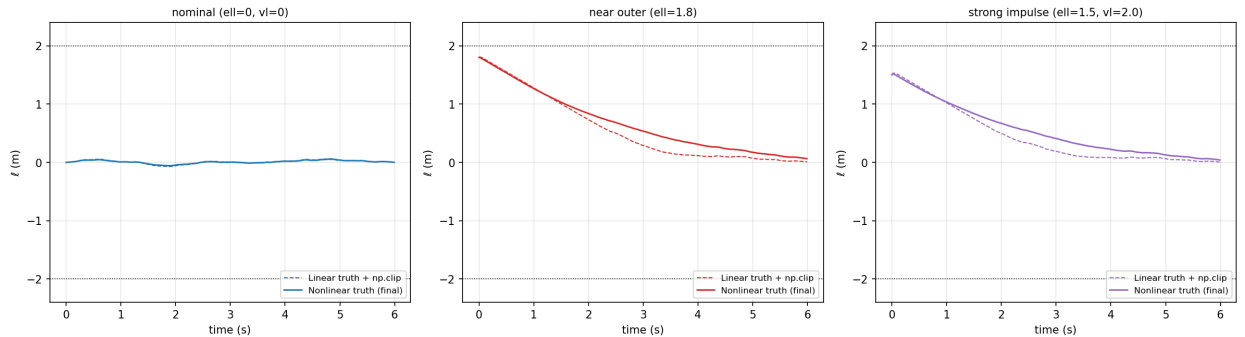
| Scenario       | $\ell_0$ (m) | $v_{\ell,0}$ (m/s) | Mean lat err (m) | Max $ \ell $ (m) |
|----------------|--------------|--------------------|------------------|------------------|
| A: nominal     | 0.0          | 0.0                | 0.0186           | 0.068            |
| B: near +bound | +1.8         | +0.3               | 0.0207           | 1.803            |
| C: near –bound | –1.8         | –0.3               | 0.0217           | 1.803            |



**Figure 11:** Near-boundary stress test under the nonlinear truth model  $f(\cdot)$  for scenarios A (centre), B (near +bound), C (near -bound). Vehicles in B and C return to the centreline within  $\sim 5$  s; steady-state EKF lateral error is at the centimetre level across all three.

All three scenarios yield steady-state lateral accuracy of 0.0186–0.0217 m. The maximum excursion under the most aggressive initial condition (scenario B) is 1.803 m, leaving a 0.197 m margin to the physical  $\pm 2$  m bound; this is consistent with the modelling-side near-outer-wall test of Section 1.2, both indicating that the nonlinear spring physically returns the vehicle from a near-boundary start. The complementary estimator-side observation is that the EKF tracks the centring transient with centimetre-level lateral error (mean  $\approx 0.02$  m in Table 6) even though the true lateral state changes by 1.8 m in the first 5 s. The slight ordering  $A < B < C$  ( $0.0186 < 0.0207 < 0.0217$  m) reflects the brief excursion through the high-stiffness region  $|\ell| \approx 1$ , where the local Jacobian  $A_t$  differs most from its nominal value  $A_0$ ; the increase is at the millimetre level and recovers within the warm-up window. Because the prediction step uses the local Jacobian rather than  $A_0$ , the filter tracks accurately even when the simulator briefly traverses the strongly nonlinear region.

**Linear vs nonlinear truth model.** Figure 12 compares the bound-respecting nonlinear truth model  $f(\cdot)$  used in this report with a linear surrogate that relies on post-hoc `np.clip` for boundary enforcement. Under nominal initial conditions the two trajectories are closely matching — in the small-deviation regime the nonlinear stiffness and tanh saturations have not activated, so the nonlinear model reduces to its linear local approximation. Near the boundary the two diverge: the linear model produces a non-physical kink at the  $\ell = \pm 2$  m clip, while the nonlinear stiffness yields a smooth  $C^\infty$  return. Under a strong adversarial impulse ( $\ell_0 = 1.5$ ,  $v_{\ell,0} = 2.0$  m/s, beyond the physical  $v_\ell$  bound) the linear-clip model also exhibits oscillations not present in the nonlinear case. The nonlinear model therefore adds value in the regime where the linear surrogate would mislead, motivating its use as the truth model throughout this report.



**Figure 12:** Linear truth model with post-hoc `np.clip` (dashed) versus bound-respecting nonlinear truth  $f(\cdot)$  (solid) under three initial conditions: nominal centre, near the outer wall, and a strong adversarial impulse beyond the physical  $v_\ell$  bound.

### 3 Comparison with Moving Horizon Estimation

A complete Moving Horizon Estimator (MHE) implementation is now presented alongside the associated real-time-feasibility analysis, establishing the EKF as the preferred estimator for this implementation for the problem-statement 100 Hz real-time requirement.

The MHE is implemented following the Bayesian full-information estimation framework presented in the course lecture notebook [9], which solves a constrained optimisation problem at each time step, minimising the Bayesian negative log-likelihood over a sliding window of  $M$  most recent measurements,

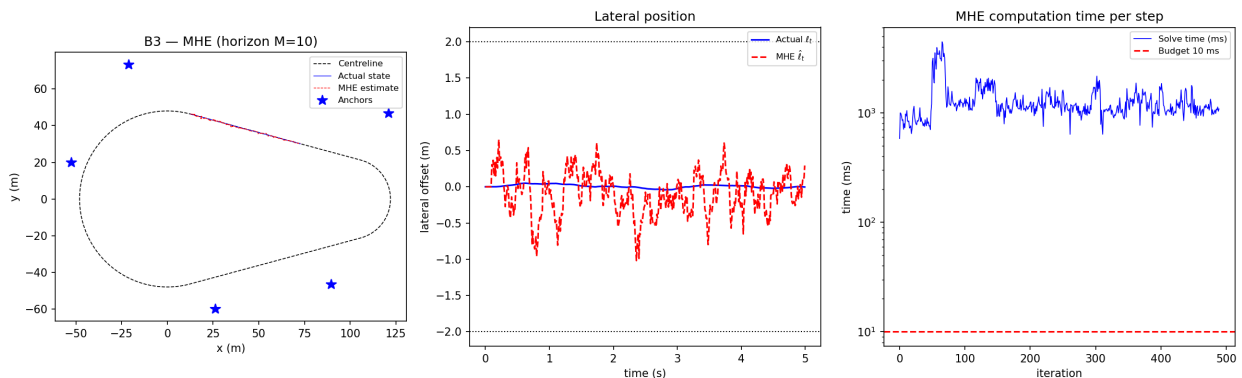
$$\begin{aligned} \min_{x_{0:M}, w_{0:M-1}} \quad & \ell_{x_0}(x_0) + \sum_{\tau=0}^{M-1} \ell_w(w_\tau) + \sum_{\tau=0}^M \ell_v(y_\tau - h(x_\tau)), \\ \text{s.t.} \quad & x_{\tau+1} = f(x_\tau, w_\tau), \quad \ell \in [-2, 2] \text{ m}, \quad v_s \in [0, 25] \text{ m/s}. \end{aligned} \quad (16)$$

The dynamics constraint is the same nonlinear  $f(\cdot)$  used by the EKF prediction step in (11). The three cost components  $\ell_{x_0}$ ,  $\ell_w$ ,  $\ell_v$  correspond to the arrival cost, process-noise penalty and measurement-residual penalty used in the course notebook. The MHE and the EKF therefore share the identical truth model and identical prediction dynamics; they differ only in the solver (constrained-optimisation versus closed-form Kalman update) and the horizon (sliding window of  $M$  vs. single time step).

The optimisation was solved using SLSQP from SciPy [5] with horizon  $M = 10$ , giving a decision variable of dimension 64 and 40 equality constraints. The choice  $Mh = 0.10$  s corresponds to roughly 1 m of vehicle motion at the nominal 10 m/s, a natural scale for a finite-lookback arc-length estimator and a deliberately short window: SLSQP cost grows super-linearly in  $M$ , so a longer horizon would only widen the real-time gap reported below. If even this short horizon is infeasible, a longer horizon would not improve real-time feasibility either. The arrival cost uses the previous MHE output as mean with a fixed diagonal prior covariance (self-bootstrapping). The constraint Jacobian  $\partial[x_{\tau+1} - f(x_\tau, w_\tau)]/\partial(x, w)$  is constructed once symbolically with CasADi and passed to SLSQP analytically rather than recomputed by finite differences at every solver iteration; this is essential for both speed and reliability under the nonlinear constraint. Results are shown in Table 7 and Figure 13.

**Table 7:** EKF versus MHE. With the analytical CasADi-derived constraint Jacobian, SLSQP converges in every one of the 490 horizon solves; mean solve time exceeds the 10 ms sampling budget by a factor of 128.

| Estimator                   | Mean arc (m) | Mean lat (m) | Solve time (ms) | Real-time at 100 Hz           |
|-----------------------------|--------------|--------------|-----------------|-------------------------------|
| EKF (this report)           | 0.226        | 0.022        | 0.43            | Yes (4.3% of budget)          |
| MHE ( $M = 10$ , nonlinear) | 0.293        | 0.229        | 1275            | No ( $128\times$ over budget) |



**Figure 13:** MHE performance and per-step solve time over the  $N = 500$  horizon. Mean solve time 1275 ms,  $128\times$  over the 10 ms sampling budget (red dashed); zero SLSQP failures in 490 solves.

The MHE achieves comparable arc-length accuracy but a larger lateral error (0.229 m versus the EKF’s 0.022 m). This larger lateral error should not be read as a fundamental limitation of MHE: it is most likely an artefact of implementation choices made for computational tractability, namely the short  $M = 10$  horizon, a fixed diagonal arrival-cost covariance rather than a Kalman-derived arrival covariance, and a local SLSQP solve. A longer horizon or an arrival cost propagated from the EKF covariance would be expected to narrow the gap, but at the cost of further increasing runtime. More fundamentally, the mean MHE solve time of 1275 ms exceeds the 10 ms sampling budget by a factor of 128, making MHE incompatible with real-time operation at 100 Hz on this hardware. The EKF, in contrast, completes in 0.43 ms (Section 5). The comparison therefore supports the practical conclusion that this implemented MHE is not real-time feasible on this problem, rather than a general claim that MHE is less accurate than EKF.

## 4 Bias-Augmented EKF: Implementation and Verification

The problem statement requires a discussion of how the estimator handles sensor bias. Section 1.5 derives the theoretical formulation: an augmented state  $z_{\text{aug}} = [s, \ell, v_s, v_\ell, \beta_1, \dots, \beta_5]^\top \in \mathbb{R}^9$  with block-diagonal  $A_{\text{aug}}, G_{\text{aug}}, Q_{\text{aug}}$  matrices, following the state-augmentation approach of [8, 10] and [2] (Ch. 13). The detectability analysis ( $\text{rank}(\mathcal{O}) = 8$  with  $e_{v_\ell} \notin \text{null}(\mathcal{O})$ ) and the noise-free measurement-function verification (recovery to within  $6 \times 10^{-15}$  m) are also developed in Section 1.5.

The present section provides the corresponding noise-domain verification: the 9-state augmented EKF is implemented and run on the simulator output with realistic measurement noise ( $\sigma_y = 1.5$  m) and injected biases. Section 1.5 sets up the augmented-state formulation; this section verifies its engineering viability under noise.

The augmented measurement function is

$$h_{\text{aug},i}(z_{\text{aug}}) = \|p(s, \ell) - a_i\|_2 + \beta_i + v_i, \quad i = 1, \dots, 5, \quad (17)$$

where  $a_i$  is the known beacon position and  $\beta_i$  is the unknown bias to be estimated. All Jacobians  $\partial h_{\text{aug}}/\partial z_{\text{aug}}$  and  $\partial h_{\text{aug}}/\partial v$  are built symbolically in CasADi; the bias sub-Jacobian  $I_5$  emerges automatically from the additive structure of (17), with no hand coding required. The augmented measurement-update and time-update functions follow the same pattern as the baseline 4-state primitives.

To verify the augmented filter under noise, a 43 s single-lap simulation was run with the bias vector

$$\beta = [+0.5, -0.3, +0.2, +0.8, -0.4]^\top \text{ m}$$

injected on the five beacons (the same test vector used in Section 1.5 for the symbolic  $h_{\text{aug}}$  verification). Two filters were run in parallel on the identical biased measurement stream: the 9-state augmented EKF, and the 4-state EKF of Section 2 (baseline that ignores bias). Results are in Table 8 and Figure 14.

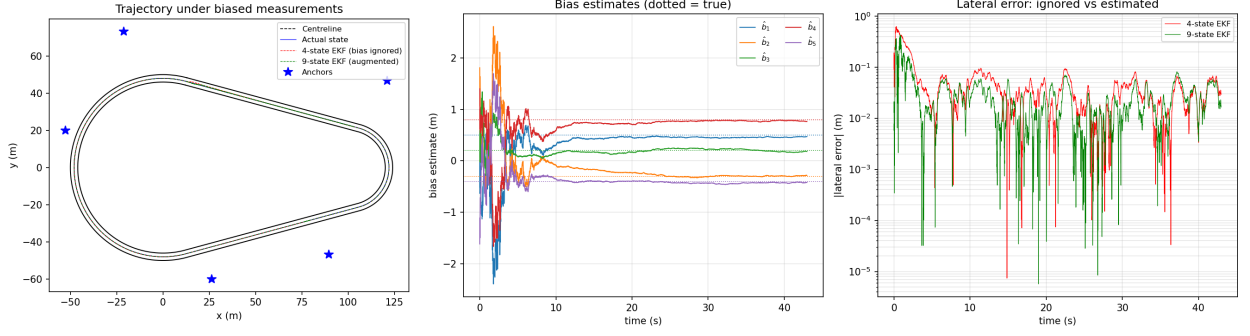
**Table 8:** Bias recovery (top) and trajectory accuracy (bottom). All five biases recovered to within 0.0354 m; the augmented filter reduces mean arc-length error by a factor of 2.1.

| Beacon    | True bias (m) | Estimated (m) | Error (m) |
|-----------|---------------|---------------|-----------|
| $\beta_1$ | +0.500        | +0.473        | −0.0270   |
| $\beta_2$ | −0.300        | −0.282        | +0.0176   |
| $\beta_3$ | +0.200        | +0.183        | −0.0167   |
| $\beta_4$ | +0.800        | +0.765        | −0.0354   |
| $\beta_5$ | −0.400        | −0.419        | −0.0195   |

| Filter                       | Mean arc-length error (m) | Mean lateral error (m) |
|------------------------------|---------------------------|------------------------|
| 4-state EKF (bias ignored)   | 0.510                     | 0.0334                 |
| 9-state EKF (bias estimated) | 0.239                     | 0.0234                 |





**Figure 14:** 9-state augmented EKF (green) versus 4-state baseline (red) under injected beacon biases. Bias estimates converge to the true values within  $\sim 5$  s; lateral error is reduced relative to the bias-ignoring baseline.

All five biases are recovered to within  $0.0354$  m, well below the  $\sigma_y = 1.5$  m measurement noise. Bias estimates converge to their true values within 5–10 s. The augmented filter reduces the mean arc-length error from  $0.510$  m to  $0.239$  m, a factor of 2.1 improvement, confirming that the state-augmentation approach effectively decouples bias estimation from vehicle state estimation. Together with the conservative observability discussion of Section 1.5, this provides an end-to-end treatment of the bias-handling requirement (theoretical formulation plus verified noise-domain implementation), rather than a theoretical discussion alone.

**Sensitivity to the bias-drift variance  $q_\beta$ .** The bias dynamics use a random-walk model with drift variance  $q_\beta = 10^{-6}$  m<sup>2</sup>/step. Because this is a tuning parameter rather than a measured quantity, a four-decade scan was run to verify that bias recovery is not sensitive to the exact choice. Table 9 shows that  $\max |\hat{\beta}_i - \beta_i|$  varies smoothly between  $0.083$  m at  $q_\beta = 10^{-8}$  and  $0.118$  m at  $q_\beta = 10^{-5}$ , with the baseline value  $q_\beta = 10^{-6}$  yielding  $0.091$  m. The vehicle’s lateral accuracy is essentially unaffected ( $0.023$  m across the scan, identical to the no-bias baseline of Section 2.3). The numbers in this scan are higher than the  $0.0354$  m of Table 8 because the scan uses a shorter  $N = 2000$  step run for budget reasons; the trend across  $q_\beta$  values is what matters.

**Table 9:** Sensitivity of bias recovery to the bias-drift variance  $q_\beta$ .  $N = 2000$  steps per run, warm-up 500 steps.

| $q_\beta$ (m <sup>2</sup> /step)     | $10^{-8}$ | $10^{-7}$ | $10^{-6}$ | $10^{-5}$ |
|--------------------------------------|-----------|-----------|-----------|-----------|
| $\max  \hat{\beta}_i - \beta_i $ (m) | 0.083     | 0.084     | 0.091     | 0.118     |
| Mean lat err (m)                     | 0.024     | 0.024     | 0.024     | 0.023     |

## 5 Computational Complexity and Real-time Feasibility

The computational cost of the EKF measurement update is dominated by the product  $C \Sigma C^\top$  and the linear solve of the  $5 \times 5$  innovation matrix  $S$ , which scale as  $O(n_y^2 n_x)$  and  $O(n_y^3)$  respectively, with  $n_x = 4$  and  $n_y = 5$ . The prediction step adds two CasADi function evaluations per step (for  $A_t$  and  $L_t$ ) and one matrix product  $A_t \Sigma A_t^\top$ . For the 9-state augmented EKF, the state-side cost rises modestly to  $O(n_y^2 n_{\text{aug}})$  with  $n_{\text{aug}} = 9$ . Each operation involves only small dense matrices and one CasADi call.

A direct `timeit` benchmark of the two 4-state EKF primitives over 500 repetitions in the setup-cell environment of `Estimator.ipynb` gives mean per-call times of  $t_{\text{meas}} = 0.20$  ms and  $t_{\text{time}} = 0.23$  ms, for a combined per-step EKF runtime of  $0.43$  ms (4.3% of the 10 ms sampling budget). The 9-state augmented variant has not been benchmarked separately here; based on the  $O(n_y^2 n_{\text{aug}})$  scaling above and the small additional CasADi calls per step, it is expected to remain inside the 10 ms budget, but this should be confirmed by a direct measurement before any embedded deployment. The 4-state benchmark is exported to



results\_summary.json so any drift between the reported runtime and the implementation is detectable on every notebook execution.

## 6 Conclusion

A vehicle state estimator for an asymmetric teardrop track was designed and validated, combining a bound-respecting nonlinear modelling component with a CasADi-based EKF implementation. The modelling chapter formulated a smooth  $C^\infty$  truth model that enforces the lane-corridor and velocity constraints by structural design rather than post-hoc clipping, derived a five-beacon GDOP-minimising layout that attenuates ranging noise across 100% of the track, validated the noise-parameter choices by Monte-Carlo scan, and developed the state-augmentation formulation used for sensor-bias handling. The estimator chapter implemented the EKF with prediction-step Jacobians  $A_t$  and  $L_t$  obtained from CasADi automatic differentiation at every step from the same dynamics that serve as the truth simulator. Across 100 Monte-Carlo trials the filter achieved a mean arc-length error of 0.226 m, well below the single-measurement  $\sigma_y \overline{\text{GDOP}} = 1.379$  m scale reference; it passed a Normalized Innovation Squared consistency check (empirical mean 4.994, 95% of samples in the 95% band); and its closed-loop covariance trace decayed by approximately  $5\times$  over the warm-up window. A Moving Horizon Estimator with the same nonlinear constraint exceeded the 10 ms budget by  $128\times$ , supporting the EKF as the preferred estimator. The 9-state augmented variant recovered injected beacon biases up to 0.8 m to within 0.04 m, with sensitivity to the bias-drift variance remaining below 0.12 m across four decades. The baseline 4-state EKF pipeline executes in 0.43 ms per step.

**Working assumptions and limitations.** The problem statement specifies only the range standard deviation  $\sigma_y = 1.5$  m and not the full probability density of the ranging error. The EKF, NIS analysis and MHE arrival cost in this report all use a Gaussian working model on  $v$ , which is standard for covariance-based Kalman filtering but remains an assumption: heavy-tailed multipath or mixture-noise statistics would call for robust alternatives such as innovation gating, Huber losses, or particle / robust filtering. Verifying robustness under non-Gaussian range noise is left for future work.

**Future work.** Opportunities for further work, not pursued here due to scope constraints, include: (a) combining the EKF and MHE into a warm-start hybrid where the EKF provides the arrival cost for a short-horizon MHE invoked only near the track boundary; (b) replacing the SLSQP solver with a code-generated QP solver tailored to the banded horizon structure, which could close the real-time gap by one to two orders of magnitude; (c) extending the bias model from a pure random walk to a first-order Markov process, accommodating slowly varying drift (e.g. thermal) in addition to constant offset; and (d) testing robustness under heavy-tailed or mixture range-noise distributions.

## AI Usage Statement

Claude (Anthropic), Gemini (Google) and ChatGPT (OpenAI) were used only as auxiliary tools, principally for wording,  $\text{\LaTeX}$  formatting, figure-caption polishing, and consistency checking between the notebooks and the report. All modelling decisions, estimator implementation, simulations, numerical results, debugging and final technical judgements were made and verified by the authors; every number in the report is regenerated from the code on each run and exported to results\_summary.json. AI-generated suggestions were not accepted without verification: in particular, an initial LLM-produced description of static-rank observability that overstated detectability was rewritten conservatively (Section 1.5).

## Appendix A: Project Management and Collaboration

### A.1 Meeting minutes

Table 10 summarises the team’s progress between the kick-off (24 March 2026) and final submission. All meetings were attended by Chenhui Cui and Weiting Wu unless otherwise indicated; commit hashes refer to the GitHub repository in A.2.

**Table 10:** Summary of meeting minutes.

| Date        | Mode             | Work completed                                                                                                                                                   | Follow-up actions                                                                                             |
|-------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| 24 Mar      | in person        | Kick-off; agreed on Python/Jupyter/CasADi workflow and repository structure.                                                                                     | Chenhui: track geometry and state-space draft. Weiting: EKF practice and estimator skeleton.                  |
| 30 Mar      | online           | A1–A3 and B1 first pass complete.                                                                                                                                | Chenhui: A4–A5 (noise scan, sensor-bias augmentation). Weiting: B2–B3 (Monte-Carlo, MHE).                     |
| 6 Apr       | online           | A4/A5/B2/B3 complete. MHE found not real-time feasible.                                                                                                          | Chenhui: A6 modelling chapter. Weiting: B4 estimator chapter.                                                 |
| 19 Apr      | online           | Estimator code had not yet propagated Chenhui Cui’s 1 April modelling rewrite (commit 590cf6b: corrected $R_B = 22$ m and re-optimised five-beacon coordinates). | Weiting Wu: correct Step 4 (commit b037600) and re-run B1–B3.                                                 |
| 21 Apr      | online           | Chenhui Cui: bound-respecting nonlinear model and matching $Q$ update (commit 495da79). Weiting Wu: further $R_B$ /EKF refresh (commit ee33948).                 | Weiting Wu: import $f_{\text{dynamics}}$ into the estimator notebook.                                         |
| 24 Apr      | in person        | Chenhui Cui consolidated PR #2 and merged Weiting Wu’s PR #4 and PR #5.                                                                                          | Chenhui Cui: create .mailmap (commit 1bbb644) and AUTHORS.md; rename the modelling notebook (commit 8b88af4). |
| 25 Apr (am) | online           | Weiting Wu: PR #6 (beacon-count ablation), PR #7 (boundary stress test), and NIS check (commit 0bb2c33); all built on Chenhui Cui’s modelling.                   | Modelling-side code freeze.                                                                                   |
| 25 Apr (pm) | Chenhui Cui only | Final consolidated estimator notebook (16 unified patches); all figures regenerated; results_summary.json exported.                                              | Integrated final report; final proofreading.                                                                  |

### A.2 Version control and collaboration

The team used git and GitHub for version control: [github.com/cuichenhui9-eng/ELE8101-CW2](https://github.com/cuichenhui9-eng/ELE8101-CW2). Modelling work was developed on the ChenhuiCui branch, estimator work on the WeitingWu branch, and final integration was merged into main. Tasks were tracked through GitHub Issues and Pull Requests, with the modelling-side polish reviewed in Pull Request #2 and the estimator-side milestones tracked in Issue #3.

### A.3 Individual contributions

**Chenhui Cui.** Chenhui Cui led the modelling component: track geometry; the bound-respecting nonlinear vehicle motion model; GDOP-based five-beacon optimisation; noise-parameter validation; and the state-augmentation formulation for sensor bias. Chenhui Cui wrote the final versions of Section 1, Sections 4–6, and the appendix, and carried out the final integration pass. At the report level, Chenhui Cui integrated and

substantially revised Weiting Wu’s initial EKF/MHE drafts into the final Sections 2–3, and was responsible for the overall report structure, technical narrative organisation, figure/table placement, notation standardisation, formatting and final manuscript polish. During the code-integration pass the estimator notebook was synchronised with the nonlinear modelling pipeline: the EKF prediction step was updated to use per-step CasADi-derived Jacobians ( $A_t$ ,  $L_t$ ); the truth simulator was unified across the validation cells; the MHE constraint was given an analytical Jacobian; closed-loop covariance and the  $P_0 / q_\beta$  sensitivity analyses were added; and a master-seed reproducibility chain together with a JSON results-summary export was introduced. The final estimator figures and numerical tables were regenerated from this integrated implementation.

**Weiting Wu.** Weiting Wu developed the original estimator workflow, including the first-pass EKF single-run simulation, the Monte-Carlo validation, the NIS consistency check, the MHE comparison, the beacon-count ablation, and the near-boundary stress test. He also provided the initial estimator-side write-up for the EKF and MHE comparison sections. These contributions provided the baseline estimator structure and validation workflow that were later integrated with the final nonlinear modelling pipeline; the corresponding commits remain on the WeitingWu branch.

**Joint work.** Both members contributed to defining the modelling–estimator interface, reviewing assumptions after the geometry correction, checking simulation outputs, selecting the report figures, revising the abstract and conclusion, reviewing the final report before submission, and preparing the repository documentation (README.md, AUTHORS.md, .mailmap).

#### A.4 Team reflection

The collaboration worked because the modelling/estimator interface was defined early, allowing the two members to develop their notebooks in parallel without integration friction. The principal challenge was synchronising parameters after the mid-project corrections to the small-arc radius and the move to bound-respecting dynamics; reproducibility across the two notebooks initially relied on each notebook calling `np.random.seed(42)` independently, which left the global RNG state fragile. The final integration round replaced this with a single MASTER\_SEED feeding per-cell `numpy.random.default_rng` instances, regenerated every figure under the corrected geometry, and cross-checked all reported numbers against the auto-exported `results_summary.json` file.

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